

Joshua (Hyun Jo) Jang

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Portfolio: musicofmusix.github.io | Github: github.com/musicofmusix

Research Interests

Computational Photography, Digital Image Processing, Image Aesthetics, Graphic Design.

Education

University of Toronto. 2020 - Present. HBSc in Computer Science and Cognitive Science

- Graduating June 2026
 - Cumulative GPA: 3.89/4.00 (Computer Science 3.97)
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Experience

AI Center - Toronto, Samsung Electronics. January 2025 - May 2025. Research Internship

- Supervisor: Prof. Michael Brown, Dr. Hakki Karaimer
 - Implemented a suite of Python tools for in-camera processing (ISP) analysis, including a RAW DNG renderer, a colour chart patch extractor, and various colour conversions.
 - Captured RAW-JPG image pairs in various environments and lighting for ISP datasets.
 - Developed a simple, explainable model for photographic film colour (See **Publications**).
 - Devised a methodology for analysing various proprietary ISP routines with a well-calibrated monitor (See **Patents**).
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Publications

Hyun Jo Jang, Hakki Karaimer, Michael Brown. Emulating Emulsion: A Compact Physically-Based Model for Film Colour. **SIGGRAPH Posters 2025**. [Project Page](#)

Patents

Hakki Karaimer, **Hyun Jo Jang**, Michael Brown. In-Camera Processing Analysis with a High-End Digital Monitor*. **Worldwide Patent Application**, 2025. Samsung Electronics Co Ltd.

**Title subject to modification.*

Creative Projects

Aesthetics-Driven Game UI Demo. 2022. Unity, Blender 3D, C#. [Project Page](#)

- Blended the visual flair of 3D motion graphics and the interactivity of video game UI
- Custom-made cartoon-style non-photorealistic rendering (NPR) shaders
- Implemented random-generated Bezier splines for stylistic effects

Interactive Virtual Subway Station. 2023. Unity, Blender 3D, Kinect, C#. [Project Page](#)

- Installed as part of an interactive media exhibition in Gimpo, South Korea
- Virtual subway would arrive & depart upon visitors entering the exhibition space

Photography, Graphic Design, 3D Modelling & Rendering. 2020 - Present

- Fujifilm 2024 Workshop Toronto: Fisheye Lens Portraits. [Images](#)
- Poster design for SIGGRAPH 2025 Posters. [PDF](#)
- "Neo Ziggurat" 3D Art. [Images](#)
- More in [Portfolio](#)

Art Exhibitions

People Category Winner. Hart House Camera Club Annual Exhibition. 2024. [Images](#)

Skills

Programming/Scripting: Python, C, C#

2D Tools: Adobe Photoshop, Adobe Illustrator, Adobe Lightroom

3D Tools: Blender 3D (Modelling, Rendering), Maxon Cinema 4D (Rendering)

Game Engines: Unity

Languages: English (Fluent), Korean (Fluent)

#CV